Postdoctoral Research Associate - Fungal Systems and Genetics

Oak Ridge National Laboratory · Oak Ridge, TN

Job ID (this archive)	ENERGY-087
Employer	Oak Ridge National Laboratory
Location(s)	Oak Ridge, TN
Career stage (archive tag)	Graduate degree / research (Postdoctoral)
Location group	Other U.S.
Majors (archive tags)	Data Science, Biomedical Engineering
Energy area	Critical minerals / synthetic biology
Pay listed	Not listed
Recruiting status when checked	Status not confirmed: the posting text was found, but current hiring was not verified. Checked Oct 26, 2026.
Source page	https://jobs.ornl.gov/job/Oak-Ridge-Postdoctoral-Research-Associate-Fungal-Systems-and-Genetics-TN-37830/1442118500/
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-058

Saved on 2026-10-26 from the employer's own careers page. The posting text below is the employer's own wording, reformatted for printing. Live listings change and close; check the source page before applying.

Job description

Requisition Id 17304

Overview:

The Biological Systems Design Group in the Biosciences Division at Oak Ridge National Laboratory (ORNL) is seeking a Postdoctoral Researcher with expertise in fungal systems and genetics. The successful candidate will be part of the Synthetic Biology section (https://www.ornl.gov/group/synthetic-biology) and will contribute to a Laboratory Directed Research and Development (LDRD) project focused on developing bio-enabled approaches for critical mineral accessibility and recovery.

In this position, you will investigate the genetic and molecular mechanisms underlying fungal interactions with lithium-bearing minerals and other critical-material systems. Research will emphasize genetically tractable non-model fungal organisms, including the development and application of genome engineering approaches to identify and functionally characterize genes and pathways associated with mineral interactions, stress tolerance, elemental partitioning, and related phenotypes. The work will integrate fungal genetics, microbiology, molecular biology, functional genomics, and phenotypic characterization.

You will work as part of an interdisciplinary team of scientists and technical staff at ORNL and collaborate with researchers spanning biological, chemical, and materials sciences. Responsibilities will include designing and executing laboratory experiments, developing and applying genetic tools for fungal systems, analyzing quantitative and multi-omics data, interpreting and communicating research findings, and preparing results for publication and presentation to the scientific community.

Major Duties/Responsibilities:

- Perform research in microbial systems and genetics to investigate and engineer biological functions.
- Work with a diverse team of scientists.
- Interface directly with internal and external partners in various disciplines.
- Present research results at national scientific meetings.
- Publish scientific results in peer-reviewed journals in a timely manner.
- Participate in project planning and execution.
- Work with staff to develop new proposals.

- Maintain detailed and accurate records.
- Acquire and analyze data in keeping with project scheduling.
- Take a position of leadership for day-to-day laboratory operations while addressing all of ORNL's environmental health and safety requirements.
- Self-assess experimental research skill set and identify weaknesses and secure learning resources to address these issues.
- Ensure compliance with environmental, safety, health and quality program requirements.
- Maintain strong dedication to the implementation and perpetuation of values and ethics.

Basic Qualifications:

- A Ph.D. in fungal genetics, microbiology, mycology, molecular biology, or a related field is required.
- Applicants cannot have received the most recent degree more than five years prior to the date of application and must complete all degree requirements before starting their appointment.

Preferred Qualifications:

- Experience with manipulating filamentous fungi.
- Experience with developing gene editing tools for non-domesticated fungal species.
- Experience with synthetic biology tools for fungi.
- Experience with designing vector for versatile gene manipulation in fungi.
- Experience generating and analyzing large datasets.
- Effective writing and presentation skills, demonstrated by a strong publication record from previous research.
- Desire to work both independently and collaboratively as part of a multidisciplinary team.
- Proven publication record.

For additional information, contact Dr. Joanna Tannous (https://www.ornl.gov/staff-profile/joanna-tannous).

Special Requirements:

Applicants cannot have received their Ph.D. more than five years prior to the date of application and must complete all degree requirements before starting their appointment. The appointment length will be up to 24 months with the potential for extension. Initial appointments and extensions are subject to performance and availability of funding.

- This is a temporary 24-month position. The appointment length will be up to 24 months with the potential for extension. Initial appointments and extensions are subject to performance and availability of funding.
- For employment at Oak Ridge National Laboratory (ORNL), a Real ID compliant form of identification will be required. Additionally, ORNL is subject to Department of Energy (DOE) access restrictions. All employees must also be able to obtain and maintain a federal Personal Identity Verification (PIV) card as mandated by Homeland Security Presidential Directive 12 (HSPD-12) and Department of Energy (DOE) Order 473.1A, which requires a favorable post-employment background investigation.
- To obtain this credential, new employees must successfully complete and pass a Federal Tier 1 background check investigation. This investigation includes a declaration of illegal drug activities, including use, supply, possession, or manufacture within the last year. This includes marijuana and cannabis derivatives, which are still considered illegal under federal law, regardless of state laws.
- For foreign national candidates: If you have not resided in the U.S. for three consecutive years, you are not eligible for the PIV credential and instead will need to obtain a favorable Local Site Specific Only (LSSO) risk determination to maintain employment. Once you meet the three-year residency requirement, you will be required to obtain a PIV credential to maintain employment.

Please submit three letters of reference when applying to this position. You may upload these directly to your application or have them sent to Postdocrecruitment@ornl.gov with the position title and number referenced in the subject line.

Instructions to upload documents to your candidate profile:

- Login to your account via jobs.ornl.gov
- View Profile
- Under the My Documents section, select Add a Document

About ORNL:

As a U.S. Department of Energy (DOE) Office of Science national laboratory, ORNL has an impressive 80-year legacy of addressing the nation’s most pressing challenges. Our team is made up of over 7,000 dedicated and innovative individuals! Our

goal is to create an environment where a variety of perspectives and backgrounds are valued, ensuring ORNL is known as a top choice for employment. These principles are essential for supporting our broader mission to drive scientific breakthroughs and translate them into solutions for energy, environmental, and security challenges facing the nation.

ORNL offers competitive pay and benefits programs to attract and retain individuals who demonstrate exceptional work behaviors. The laboratory provides a range of employee benefits, including medical and retirement plans and flexible work hours, to support the well-being of you and your family. Employee amenities such as on-site fitness, banking, and cafeteria facilities are also available for added convenience.

Other benefits include the following: Prescription Drug Plan, Dental Plan, Vision Plan, 401(k) Retirement Plan, Contributory Pension Plan, Life Insurance, Disability Benefits, Generous Vacation and Holidays, Parental Leave, Legal Insurance with Identity Theft Protection, Employee Assistance Plan, Flexible Spending Accounts, Health Savings Accounts, Wellness Programs, Educational Assistance, Relocation Assistance, and Employee Discounts.

This position will remain open for a minimum of 5 days after which it will close when a qualified candidate is identified and/or hired.

We accept Word (.doc, .docx), Adobe (unsecured .pdf), Rich Text Format (.rtf), and HTML (.htm, .html) up to 5MB in size. Resumes from third party vendors will not be accepted; these resumes will be deleted and the candidates submitted will not be considered for employment.

If you have trouble applying for a position, please email ORNLRecruiting@ornl.gov.

ORNL is an equal opportunity employer. All qualified applicants, including individuals with disabilities and protected veterans, are encouraged to apply. UT-Battelle is an E-Verify employer.

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official posting description retrieved; application acceptance not independently established. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: A Ph.D. in fungal genetics, microbiology, mycology, molecular biology, or a related field is required.

Key skills listed in the archive: genetics; microbiology

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Unverified · PDF: Complete · Source: Official employer careers Employer / ATS identifier: 17304

Career stage: Postdoctoral

Major fit: Adjacent research pathway: fungal genetics and bio-enabled mineral recovery; related PhD required. Energy-sector connection: Biological recovery of lithium and other critical materials for energy technologies.

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Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer careers. Source identifier: 17304.

Sources and retrieval details

Employer posting

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Retrieved: 2026-10-26T16:37:10.147797+00:00 (UTC). Major fit, career stage and energy-sector connection are archive annotations; the employer text above is preserved from the cited source.